PRACTICE SHEET

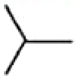
EXERCISE-I

(Alkanes and Cycloalkanes : Preparation, properties)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. Which of the following can be prepared in good yield by Wurtz reaction?

1)  2)  3)  4) 

2. Which of the following alkanes may be synthesized from a single alkyl halide by a process involving coupling reaction?

1) 2-Methylbutane 2) 2-Methylpropane
3) 2, 3-Dimethylbutane 4) Propane

3. When bromoethane is subjected to Wurtz reaction,the hydrocarbon mixture so obtained consists of

1) butane only 2) butane and ethane
3) butane and ethene 4) butane, ethane and ethene

4. The compound which produces propane on heating with HI in presence of red phosphorus is

1) $CH_3CH_2CH_2I$ 2) CH_3CH_2CHO
3) $CH_3CH_2CH_2OH$ 4) All the three

5. Propane can be best prepared by the reaction

1) $CH_3CH_2I + CH_3I \xrightarrow[\Delta]{Na/ether}$ 2) $CH_3CH_2Br + (CH_3)_2CuLi \xrightarrow[\Delta]{ether}$
3) $CH_3COONa + CH_3COONa \xrightarrow[Electrolysis]{H_2O}$ 4) $CH_3CH_2CH_2Na \xrightarrow[630K]{NaOH/CaO}$

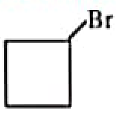
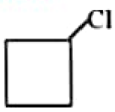
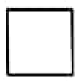
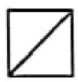
6. Which of the following alkyl halides is not suitable for Corey-House synthesis of alkanes?

1) CH_3Br 2) C_2H_5I 3) $CH_3CH_2CH_2I$ 4) $(CH_3)_3CBr$

7. Which of the following methods cannot be used to prepare propane?

1) Reaction of isopropylmagnesium bromide with water
2) Reaction of n-propylmagnesium bromide with methanol
3) Reduction of Propene with $LiAlH_4$
4) Reduction of Propyne with H_2/Raney Ni

8. 1-Bromo-3-chlorocyclobutane is treated with two equivalents of Na, in the presence of either. Which of the following will be formed?

1)  2)  3)  4) 

9. Which of the following alkylbromide will react with sodium to form 4,5-diethyloctane?

1) 1-Bromobutane 2) 1-Bromohexane
3) 2-Bromohexane 4) 3-Bromohexane

10. Arrange the fallowing alkanes in decreasing order of their heat of combustion:

(X) (Y) (Z)

1) X > Y > Z 2) Z > X > Y 3) Z > Y > X 4) X > Z > Y

11. An alkane cannot be chlorinated by using which of the fallowing reagents?

1) $Cl_2/h\nu$ 2) HCl 3) SO_2Cl_2 4) t-bu – O – Cl

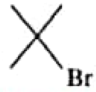
12. For the give reaction how many products will obtain (all isomers)?

1) 1 2) 6 3) 4 4) 3

13. $H_3C-\underset{CH_3}{\overset{CH_3}{C}}-H + CCl_4 \xrightarrow[h\nu]{R_2O_2}$ Product :

1) $H_3C-\underset{CH_3}{\overset{CH_3}{C}}-Cl$ 2) $CHCl_3$ 3) both (1) and (2) 4) none of these

14. Which of the fallowing alkyl halides is not suitable for Corey-house synthesis of alkanes?

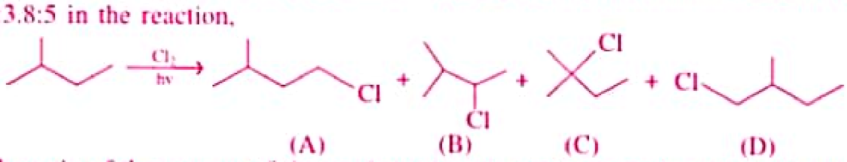
1) CH_3I 2)  3)  4) 

15. The relative reactivity of 1°H, 2°H and 3°H in bromination reaction has been found to be 1:82:1600 respectively. In the reaction,

The percentage yield of (A) and (B) are expected to be:

1) 99.4%,0.6% 2) 50%,50% 3) 0.6%,99.4% 4) 80%,20%

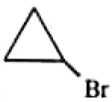
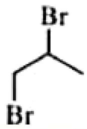
16. The relative reactivity of 1°, 2° and 3° hydrogens in chlorination reaction has been found to be 1:3.8:5 in the reaction,



The ratio of the amount of the product (A), (B), (C) and (D) is expected to be:

1) 1:3:8:5:1 2) 3:7.6:5:6 3) 3:7.6:5:3 4) 1:7.6:5:1

17.  + HBr ⟶ Product:

1)  2)  3)  4) 

EXERCISE-II

(Alkene preparation, electrophilic addition reaction, oxidation reactions, reactions of alkene)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. When 100gm Ethene polymerises entirely to poly ethene, the weight of poly ethene formed as per the equation $nCH_2=CH_2 \rightarrow (CH_2-CH_2)_n$ is

   1) $\frac{n}{2}$ g 2) 100 g 3) $\frac{100}{n}$ g 4) 100n g

2. Among the following which one is more reactive in hydrogenation

   1) 1-Butene 2) isobutene 3) Cis-2-butene 4) Trans-2-Butene

3. $CH_2=CH_2 \xrightarrow{Br_2, NaCl(Aq)}$

   In the above reaction which one of the following products is not possible?

   1) $\underset{Br}{CH_2}-\underset{Br}{CH_2}$ 2) $\underset{Br}{CH_2}-\underset{OH}{CH_2}$ 3) $\underset{Cl}{CH_2}-\underset{Br}{CH_2}$ 4) $\underset{Cl}{CH_2}-\underset{OH}{CH_2}$

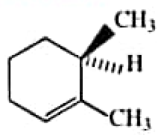
   (In each option the first substituent is on the left carbon and the second on the right carbon.)

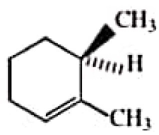
4. $(CH_3)_3C-\overset{OH}{\overset{|}{C}H}-CH_3 \xrightarrow{Conc\ H_2SO_4} X(Major) \xrightarrow[Ozonalysis]{Reductive} Y$ then Y is

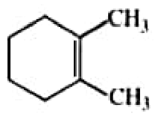
   1) Acetone 2) Methanal 3) Dimethylpropanal 4) (2) and (3)

5. 

   Correct statement among the following is

   1) X and Y are same 2) X is 

   3) Y is 

   4) Y is 

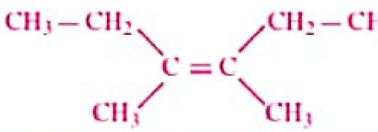
6. $CH_2=CH-CH=\underset{CH_3}{\underset{|}{C}}-CH_3 \xrightarrow{+HBr} CH_3-\overset{Br}{\overset{|}{C}H}-CH=\underset{CH_3}{\underset{|}{C}}-CH_3 + CH_3-CH=CH-\overset{Br}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}-CH_3$ (II)

   Correct statement about the above reaction is

   1) 'I' is kinetic controlled product
   2) 'II' is thermodynamic controlled product
   3) 'I' is major product at low temperature
   4) 'II' is major product at low temperature

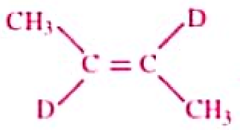
7. Among the following which one is less reactive than ethene but gives Markovnikov's product in the addition of HBr

1) Isobutene 2) $CH_2 = CH - Cl$ 3) $CH_2 = CH - NO_2$ 4) $CH_2 = CH - OCH_3$

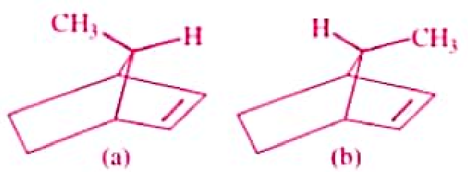
8.  $+ HBr \xrightarrow[\text{(Peroxide)}]{R_2O_2}$ Products

How many products will be formed in above reaction?

1) 2 2) 4 3) 3 4) 6

9.  $\xrightarrow[Pt]{H_2}$ Product of the reaction is

1) Racemic 2) Diastereomers 3) Meso 4) Pure enantiomers

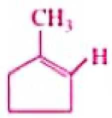
10. 

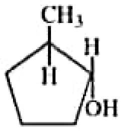
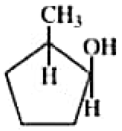
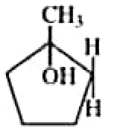
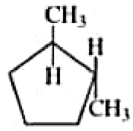
(a) (b)

Rate of reaction towards reduction using (H_2/Pt) :

1) a > b 2) a = b

3) b > a 4) Reduction of given molecule is not possible

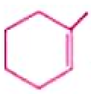
11.  $\xrightarrow[(2) H_2O_2, HO^-]{(1) BH_3; THF}$ (A); Product of the above reaction is

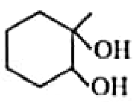
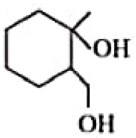
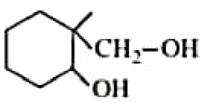
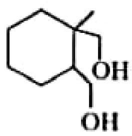
1)  2)  3)  4) 

12. $CH_3 - CH = CH_2 \xrightarrow[(2) CH_3CO_2T]{(1) THF, BD_3}$ (A) ; Product A of the above reaction is

1) $CH_3 - CHD - CH_2D$ 2) $CH_3 - CHT - CH_2T$

3) $CH_3 - CHD - CH_2T$ 4) $CH_3 - CHT - CH_2D$

13.  $\xrightarrow[H_2O]{HCHO, H^+}$ Major product of this reaction is

1)  2)  3)  4) 

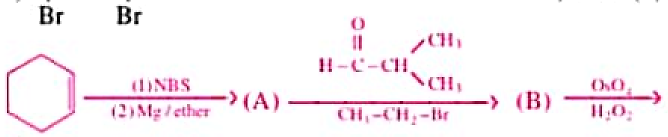
14. $CH_3-\overset{\overset{CH_3}{|}}{C}=CH-CH_3 \xrightarrow[(90\%)]{Hg(OAc)_2/EtOH} (A)$. Product (A) of the above reaction is:

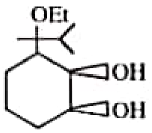
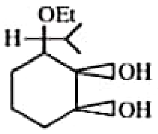
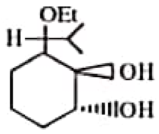
1) $CH_3-C(CH_3)(HgOAc)-CH(OEt)-CH_3$
2) $CH_3-C(CH_3)(OEt)-CH(HgOAc)-CH_3$
3) $CH_3-C(CH_3)(OH)-CH(HgOAc)-CH_3$
4) $CH_3-C(CH_3)(HgOAc)-CH(OH)-CH_3$

15. $H_2\overset{14}{C}=CH-CH_3 \xrightarrow[\text{or high temp}]{\text{low conc. of } Br_2} (?)$

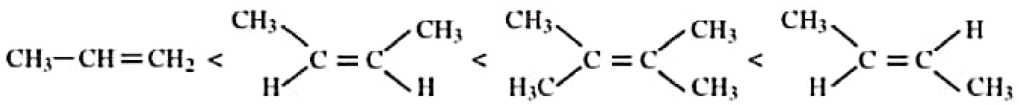
Product of the above reaction is:

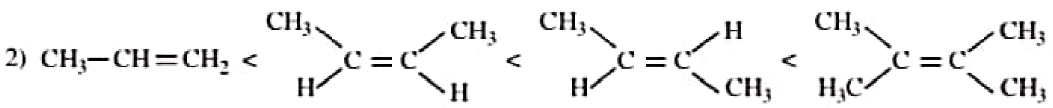
1) $H_2\overset{14}{C}=CH-CH_2-Br$
2) $H_2C=CH-\overset{14}{C}H_2-Br$
3) $\overset{14}{C}H_2(Br)-CH(Br)-CH_3$
4) both (1) and (2)

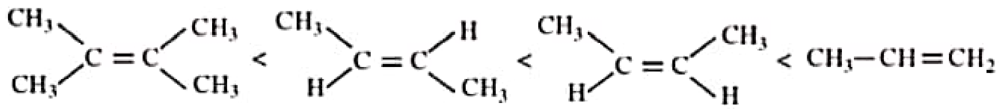
16.  (C). Product (C) is

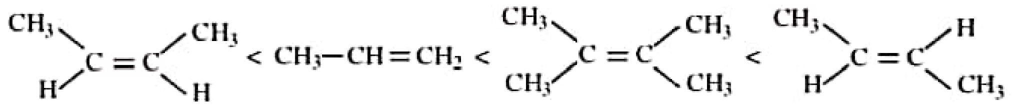
1) 
2) 
3) 
4) Both (1) and (2)

17. Which of the following is correct order of stability of alkene?

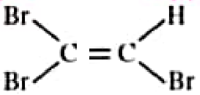
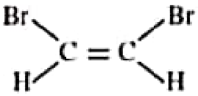
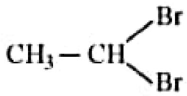
1) 

2) 

3) 

4) 

18. 

1)    2)    3) $HC \equiv CH$ 4) 

19. Bond angle in alkene (C_2H_4) is equal to

1) 120^0 2) $109^0\ 28^0$ 3) 180^0 4) 60^0

20. Lindlar's catalyst is

1) Pt in ethanol   2) pd + $BaSO_4$ 3) Ni in ethanol   4) Na in liq. NH_3

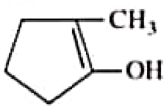
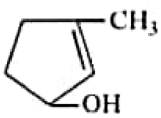
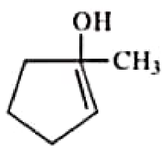
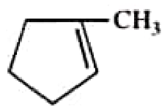
21. Reactions of alkenes with ——— are explosive in nature

1) Cl_2 2) I_2 3) Br_2 4) F_2

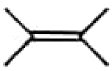
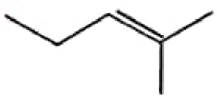
22. Which of the following molecule is not linear

1) C_2H_4 2) HCN   3) CO_2 4) C_2H_2

23.  A. A is :

1)    2)    3)    4) 

24. Rank the following in order of stability (lowest to highest).

I)    II)    III)    (IV) 

1) IV < II < III < I   2) IV < III < II < I   3) I < III < II < IV   4) IV < II = III < I

EXERCISE-III

(Alkynes preparation and general properties, alkynes reactions)

LEVEL-I (MAIN)

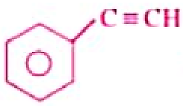
Straight Objective Type Questions

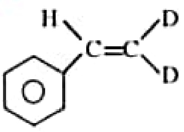
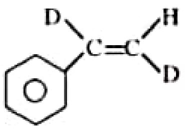
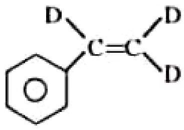
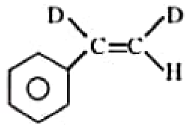
1. The end product of the following reaction is $CH_3C \equiv CH \xrightarrow{CH_3MgBr} I \xrightarrow{CH_3CH_2Br} II$

   1) $CH_3C \equiv CMgBr$ 2) $CH_3C \equiv CCH_3$ 3) $CH_3-\overset{\overset{\large CH_3}{|}}{C}=CHCH_2CH_3$ 4) $CH_3C \equiv CCH_2CH_3$

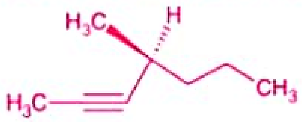
2. What is formed when calcium carbide react with heavy water?

   1) CD_2 2) CaD_2 3) CaD_2O 4) C_2D_2

3.  $\xrightarrow[Pd-BaSO_4+S]{D_2}$ A. The product (A) is

   1)  2)  3)  4) 

4. Pick out the correct statement from the following for the compound (I)

   1) On reduction in presence of poisoned Palladium catalyst it gives optically inactive compound
   2) On reduction in presence of Nickel catalyst it gives optically active compound.
   3) On reduction in presence of poisoned Palladium catalyst it gives optically active compound
   4) On reduction in Na/liquid NH_3 it gives an optically inactive compound.

5. Pick out the in-correct statement about "C" from the following for the following reaction scheme.

   $H_3C-C \equiv CH \xrightarrow{Na} A \xrightarrow{H_3C-Cl} B + H_2 \xrightarrow[\text{palladised charcoal}]{\text{partially deactivated}} C$

   1) The product C on reductive ozonolysis gives only acetaldehyde
   2) The product C can't decolourise bromine water.
   3) The product C on reaction with Br_2/CCl_4 gives racemic mixture
   4) The compound C decolourises cold alkaline $KMnO_4$ forming a meso compound

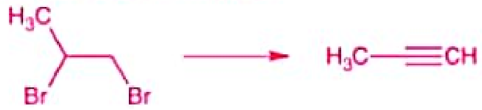
6. The reduction of 2–Butyne with H_2 in the presence of $Pd/BaSO_4$ – Quinoline and Na/liquid NH_3 gives

   1) trans -2- butene in both the cases 2) cis–2–butene in both the cases
   3) cis–2–butene and trans 2–butene 4) Butane in both the cases

7. Electrolysis of aqueous solution of sodium salt of Maleic acid and Fumaric acid gives,

   1) Ethylene and acetylene respectively 2) Acetylene and ethylene respectively
   3) Ethylene and Ethane respectively 4) Ethyne in both the cases

8. The reagent for the following reaction would be



1) Alcoholic KOH
2) Alcoholic KOH followed by $NaNH_2$
3) Zn/CH_3OH
4) Aqueous KOH followed by $NaNH_2$

9. Propyne when passed through a hot iron tube at 400^0C produces
1) benzene
2) methylbenzene
3) 1,2-dimethylbenzene
4) sym.-trimethylbezene

10. In the following sequence of reactions, $CH_3CH_2Br \xrightarrow{KOH(alc)} X \xrightarrow{Br_2} Y \xrightarrow{KOH(alc)} Z$, Z is
1) $CH_2=CH_2$
2) CH_2BrCH_2Br
3) $CH\equiv CH$
4) CH_3CH_3

11. 1,2-Dibromopropane on treatment with X moles of $NaNH_2$ followed by treatment with ethyl bromide gave a pentyne. The value of X is
1) One
2) Two
3) Three
4) Four

12. Propyne is formed by reaction of
1) CH_3Br with sodium acetylide
2) CH_3I with disodium acetylide
3) with acetyline
4) with chloroacetylene

13. 2-Butanone is heated with PCl_5 and the product so obtained is heated with alcoholic KOH, the final product obtained is
1) 1-butyne
2) 2-butyne
3) 1,2-butadiene
4) 1,3-butadiene

14. The reaction of acetylene and propyne with $HgSO_4$ in presence of H_2SO_4 produces respectively
1) acetone and acetaldehyde
2) acetaldehyde and acetone
3) propionaldehyde and acetone
4) acetone and propionaldehyde

15. Which of the following reaction will yields 2,2-dibromopropane?
1) $CH_3-CH=CH_2+HBr \rightarrow$
2) $CH_3-C\equiv CH+2HBr \rightarrow$
3) $CH_3CH=CHBr+HBr \rightarrow$
4) $CH\equiv CH+2HBr \rightarrow$

16. Predict the product C obtained in the following reaction of butyne-1
$CH_3CH_2-C\equiv CH+HCl \rightarrow B \xrightarrow{HI} C$

1) $CH_3CH_2-\overset{\displaystyle I}{\underset{\displaystyle Cl}{C}}-CH_3$

2) $CH_3-\underset{\displaystyle Cl}{CH}-CH_2CH_2I$

3) $CH_3-CH_2-CH_2-\overset{\displaystyle I}{C}-H$

4) $CH_3-CH_2-\overset{\displaystyle I}{CH}-CH_2Cl$

EXERCISE-V

(Aromaticiy and Acid-Base Properties, Aromatic Electrophilic Substitution)

LEVEL-I (MAIN)

Straight Objective Type Questions

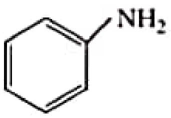
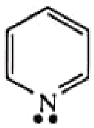
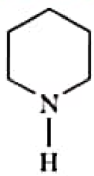
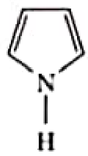
1. Aromatic compounds containing benzene are termed as

   1) Benzene 2) benzenoids 3) benzalene 4) all of these

2. The presence of delocalized $\pi-$electrons in benzene indicates that it is

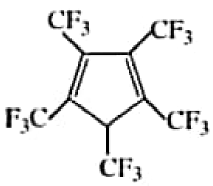
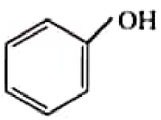
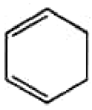
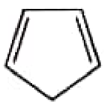
   1) less stable than cyclohexatriene 2) more stable than cyclohexatriene
   3) more basic than cyclohexatriene 4) Both (2) and (3)

12. Arrange the following compounds in increasing order of their basic strength.

I)  II)  III)  IV) 

1) I < II < III < IV 2) IV < III < II < I 3) IV < II < I < III 4) IV < II < III < I

13. What is the correct order of increasing acidic strength of the following?

I)  II)  III)  IV) 

1) III < I < IV < II 2) III < IV < II < I 3) III < II < IV < I 4) III < IV < I < II

14. In benzene molecule there are 3π-bonds and _____

1) 10 sigma bonds 2) 12 sigma bonds 3) 3 sigma bonds 4) 6 sigma bonds

15. Which of the following reaction takes place when a mixture of concentrated HNO_3 and H_2SO_4 reacts on benzene at 300 K.

1) Sulphonation 2) Nitration 3) Hydrogenation 4) Dehydration

16. Which of the following is not obtained by fractional distillation of coal tar

1) Light oil 2) Heavy oil 3) Middle oil 4) vegetable oil

17. The resonance energy of benzene is

1) 241 K.cal/mole 2) 48 K.cal/mole 3) 36 K.cal/mole 4) 76 K.cal/mole

18. Which group is meta directing?

1) $-CCl_3$ 2) $-OH$ 3) $-NH_2$ 4) $-CH_3$

1) b	2) a	3) a	4) c	5) c	6) c	7) a	8) a
9) a	10) b	11) b	12) bc	13) c	14) abc	15) ab	16) bc
17) bc	18) cd	19) abc	20) abc	21) acd	22) b	23) b	24) d
25) c	26) a	27) 5	28) 2	29) 8	30) 3		

EXERCISE-IV

LEVEL-I

1) 1	2) 3	3) 3	4) 4	5) 3	6) 1	7) 2	8) 2
9) 1	10) 2	11) 4	12) 3	13) 2	14) 4	15) 3	

LEVEL-II

1) b	2) d	3) d	4) c	5) c	6) a	7) d	8) b
9) d	10) b	11) c	12) d	13) ad	14) abc	15) acd	16) bc
17) ac	18) d	19) a	20) b	21) A-R; B-PQT; C-PQS; D-PT			
22) A-PQRS; B-PQRS; C-PQR; D-Q					23) 2		

EXERCISE-V

LEVEL-I

1) 2	2) 2	3) 1	4) 4	5) 1	6) 4	7) 4	8) 3
9) 3	10) 3	11) 3	12) 3	13) 2	14) 2	15) 2	16) 4
17) 3	18) 1						

LEVEL-II

1) d	2) b	3) c	4) b	5) d	6) a	7) b	8) d
9) c	10) d	11) c	12) b	13) d	14) ab	15) ad	16) a
17) abc	18)abcd	19) abd	20) acd	21) cd	22) bc	23) ac	24) bcd
25) c	26) d	27) a	28) 5	29) 6	30) 5	31) 3	32) 4
33) 2	34) 6	35) 6	36) 7	37) 5	38) 5		

EXERCISE-VI

LEVEL-I

1) 3	2) 2	3) 2	4) 3	5) 4	6) 2	7) 3	8) 1
9) 2	10) 2	11) 3	12) 3	13) 2	14) 3	15) 3	16) 2
17) 2							

KEY SHEET (PRACTICE SHEET)

EXERCISE-I

LEVEL-I

1) 1	2) 3	3) 4	4) 4	5) 2	6) 4	7) 3	8) 4
9) 4	10) 3	11) 2	12) 2	13) 3	14) 4	15) 1	16) 2
17) 2							

LEVEL-II

1) a	2) a	3) d	4) c	5) a	6) d	7) b	8) b
9) d	10) c	11) ab	12) acd	13) bc	14) ab	15) ab	16) acd
17) abcd	18) ab	19) bc	20) ab	21) c	22) a	23) c	24) c
25) d	26) a	27) b	28) A-Q; B-R; C-PRS; D-Q,R				

29) A-S; B-R; C-P; D-Q

EXERCISE-II

LEVEL-I

1) 2	2) 1	3) 4	4) 1	5) 3	6) 4	7) 2	8) 2
9) 1	10) 1	11) 1	12) 3	13) 2	14) 2	15) 4	16) 3
17) 2	18) 3	19) 1	20) 2	21) 4	22) 1	23) 3	24) 1

LEVEL-II

1) a	2) d	3) a	4) a	5) d	6) c	7) a	8) a
9) a	10) b	11) d	12) a	13) a	14) b	15) d	16) c
17) b	18) b	19) c	20) bc	21) acd	22) bc	23) bcd	24) abcd
25) bcd	26) bc	27) abd	28)abcd	29) acd	30) abc	31) b	32) c
33) b	34) b	35) c	36) A-S, B-R; C-PR; D-PQR				

37) A-R; B-Q; C-S; D-P

38) A-S;B-P;C-Q;D-R

39) A-PQ; B-P; C-PQS; D-R

40) A-S; B-R; C-PQ; D-PQ

41) A-R; B-PS; C-P; D-QR

42) A-QS; B-PR, C-QS; D-QS

EXERCISE-III

LEVEL-I

1) 4	2) 4	3) 4	4) 3	5) 2	6) 3	7) 4	8) 2
9) 4	10) 3	11) 3	12) 1	13) 2	14) 2	15) 2	16) 1